MUSIIWA THANDABANTU

DATA ANALYTICS

EXERCISE 01: SQL FOUNDATIONS

TABLE employees ~~department~~

id	first_name	last_name	department	salary	hire_date	city
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

Retrieve all columns from the employees table.

```
SELECT *
FROM employees;
```

01

id	first_name	last_name	department	salary	hire_date	city
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

02 Find all unique departments

```
SELECT DISTINCT (department)
FROM employees;
```

id	department
1	IT
2	HR
3	Finance
4	Marketing

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03 Retrieve first and last names ordered by salary descending

```
SELECT first_name, last_name
FROM employees
ORDER BY Salary DESC;
```

id	first_name	last_name	Salary
1	Eva	Naidoo	72000
2	Alice	Green	70000
3	Cathy	Zulu	65000
4	David	Mokoena	50000
5	Brian	Lee	45000

04 Retrieve the top 3 highest-paid employees

```
SELECT id, first_name, last_name, Salary
FROM employees
ORDER BY Salary DESC
LIMIT 3;
```

id	first_name	last_name	Salary
1	EVA	Naidoo	72000
2	ALICE	Green	70000
3	Cathy	Zulu	65000

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05 Find employees in the IT department

```
SELECT id, first_name, last_name, department
FROM employees
WHERE department = 'IT';
```

id	first_name	last_name	department
1	ALICE	Green	IT
2	EVA	Naidoo	IT

06 Find employees in Finance with Salary > 60000

```
SELECT id, first_name, last_name, department, salary
FROM employees
WHERE department = 'Finance' AND salary > 60000;
```

id	first_name	last_name	department	salary
1	Cathy	Zulu	Finance	65000

07 Find employees in HR or Marketing

```
SELECT id, first_name, last_name, department
FROM employees
WHERE department = 'HR' OR 'Marketing';
```

id	first_name	last_name	department
1	Brian	Lee	HR
2	David	Motoena	Marketing

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08. Find employees not in IT

```
SELECT id, first_name, last_name, department
FROM employees
WHERE NOT IN 'IT';
```

id	first_name	last_name	department
1	Brian	Lee	HR
2	Cathy	Zulu	Finance
3	David	Mokoena	Marketing

09 Find employees in IT, HR or Finance using IN

```
SELECT id, first_name, last_name, department
FROM employees
WHERE department IN 'IT' OR 'HR' OR 'Finance';
```

id	first_name	last_name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	Eva	Naidoo	IT

10 Find employees in IT with salary > 65000 and city Johannesburg

```
SELECT id, first_name, last_name, department, salary, city
FROM employees
WHERE department = 'IT' AND salary > 65000
AND city = 'Johannesburg';
```

id	first_name	last_name	department	salary	city
1	Alice	Green	IT	70000	Johannesburg
2	Eva	Naidoo	IT	72000	Johannesburg

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